

fs_mpc_mg — Detailed Plan for Simulation Studies

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26 April 2026

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fs_mpc_mg — Detailed Plan for Simulation Studies

Author: Anoop Eluvathingal **Date:** 26 April 2026 **Scope:** Concrete pre-HIL simulation programme for the centralized MIC + DT system. **Companion:** fs_mpc_microgrid/ (57 passing tests, ~5,500 LOC).

The Phase-1–3 work has *built* the system. This document specifies what to *measure* with it. Each study has a numbered objective, a methodology, parameter sweeps, success criteria, and a deliverable artefact (figure or table). The studies fan out from reproducibility (Class A) through coordination (Class C) to digital-twin validation (Class E), and end with the readiness gate for HIL (Class F).

1. Study taxonomy

Class	Theme	# studies	Total runs
A	Reproducibility & baseline	4	~30
B	Sensitivity & parameter sweeps	5	~120
C	Coordination (multi-ICA fleet)	4	~60
D	Faults & resilience	5	~80
E	Digital-twin validation	4	~50
F	Real-time / HIL prep	3	~20
Total		25	~360

All runs are scriptable in pure Python via the existing `scripts/` entry-points; an outer driver (`scripts/run_studies.py`, to be written) generates parameter grids and aggregates results into CSV + plots.

2. Class A — Reproducibility & Baseline

A.1 Three-mode reproduction (already complete)

Field	Value
Objective	Reproduce Perez 2016 Fig. 4–6; report THD and v_{dc} settling.
Method	<code>run_loading_mode.py</code> , <code>run_regenerating_mode.py</code> , <code>run_statcom_mode.py</code> .
Sweep	1 run per mode at nominal parameters.
Success	THD $i_s < 5\%$ in all modes; v_{dc} within 1 % of ref at $t = 80$ ms.
Artefact	<code>figures/loading_mode.png</code> etc.
Status	Complete (THD = 4.29 / 2.68 / 5.75 %).

A.2 Energy-domain vs linear- v_{dc} PI

Field	Value
Objective	Quantify the benefit of the energy-domain DC-link control (Eq. 11 of [F]) over a small-signal linearised v_{dc} PI.
Method	Implement a baseline <code>LinearVdcPI</code> next to <code>EnergyPI</code> in a side-by-side script.
Sweep	100 A i_{dc} step at three operating points (loading 80 A, regen 80 A, STATCOM).
Success	Energy PI shows $\leq 0.5x$ the overshoot and $\leq 0.75x$ the settling time of the linear PI in all cases.
Artefact	<code>figures/study_A2_energy_vs_linear_pi.png</code> , <code>study_A2_metrics.csv</code> .

A.3 Delay-compensated vs naive predictor

Field	Value
Objective	Confirm that <code>predict_with_delay_comp</code> reduces tracking error vs the single-step Eq. 4.
Method	Toggle <code>FSMPParams.use_delay_compensation</code> .
Sweep	$T_s \in \{10 \mu s, 20 \mu s, 50 \mu s\}$, three modes.
Success	Delay-comp at $T_s = 20 \mu s$ gives $\leq 0.5x$ the steady-state RMS tracking error of the naive predictor.
Artefact	<code>figures/study_A3_delay_compensation.png</code> .

A.4 6-pulse rectifier vs harmonic injector

Field	Value
Objective	Confirm that the topology-faithful <code>RectifierLoad</code> and the parametric <code>HarmonicLoad</code> produce equivalent THD when matched.
Method	Match <code>RectifierLoad</code> 's I_d to <code>HarmonicLoad</code> 's fundamental amplitude; compare $i_l(t)$ waveforms.
Sweep	Three power levels (10 / 25 / 50 kW).
Success	THD difference < 2 percentage points; phase relationships of dominant harmonics within $\pm 5^\circ$.
Artefact	<code>figures/study_A4_load_models.png</code> .

3. Class B — Sensitivity & parameter sweeps

B.1 Sample-time T_s

Field	Value
Objective	Find the largest T_s at which the FS-MPC remains stable and meets $\text{THD} < 5\%$.
Method	Loading mode reproduction with $T_s \in \{5, 10, 20, 33, 50, 75, 100\} \mu\text{s}$.
Sweep	7 values $\times$ 3 modes = 21 runs.
Success	Stability surface $\text{THD}(T_s, \text{mode})$; identify the breakdown T_s .
Artefact	<code>study_B1_Ts_surface.png</code> .

B.2 Filter inductance L drift

Field	Value
Objective	Robustness of FS-MPC predictor to L mismatch between controller and plant.
Method	Plant $L_{\text{true}} \in \{0.7, 1.0, 1.3, 1.5\} \text{ mH}$ while controller assumes 1.0 mH.
Sweep	4 values $\times$ 3 modes = 12 runs.
Success	$\text{THD } i_s \leq 6\%$ across the range; settling time variation $< 30\%$.
Artefact	<code>study_B2_L_robustness.png</code> .

B.3 DC capacitance C drift

Field	Value
Objective	Effect of C ageing on outer-loop bandwidth and overshoot.
Method	Plant $C_{\text{true}} \in \{500, 750, 1000, 1500\} \mu\text{F}$; controller assumes 1000 μF .
Sweep	4 values $\times$ 3 modes = 12 runs.
Success	Overshoot $\Delta < 10\%$ of $v_{\text{dc_ref}}$; no instability.
Artefact	<code>study_B3_C_robustness.png</code> .

B.4 Grid-impedance / SCR sensitivity

Field	Value
Objective	Performance under weak-grid conditions (low short-circuit ratio).
Method	Add Thevenin impedance $Z_{\text{grid}} \in \{0, 0.05, 0.10, 0.20\} \text{ pu}$ in series with v_s .
Sweep	4 impedances $\times$ 3 modes $\times$ 2 PLLs (Ideal vs SOGI) = 24 runs.
Success	SOGI-PLL maintains lock with Z_{grid} up to 0.20 pu; THD remains $\leq 6\%$.
Artefact	<code>study_B4_weak_grid.png</code> .

B.5 PLL bandwidth ω_n

Field	Value
Objective	Map ω_n vs lock time vs harmonic immunity.
Method	SOGI-PLL $\omega_n \in \{50, 100, 200, 400, 800\}$ rad/s; inject 5 % 5th + 5 % 7th.
Sweep	$5 \omega_n \times 3$ distortion levels = 15 runs.
Success	Pareto front identified; thesis-recommended setting documented.
Artefact	<code>study_B5_PLL_pareto.png</code> .

4. Class C — Coordination (multi-ICA fleet)

C.1 Fleet size scaling

Field	Value
Objective	THD and dispatch latency as fleet grows.
Method	$N \in \{1, 2, 4, 8, 16\}$ ICAs; total load held at 100 kW.
Sweep	5 fleet sizes $\times$ 3 modes = 15 runs.
Success	Per-ICA THD $i_s \leq 5$ %; aggregate PCC THD ≤ 3 %.
Artefact	<code>study_C1_fleet_scaling.png</code> , csv of dispatch latencies.

C.2 Q-allocation policy comparison

Field	Value
Objective	Closed-form proportional vs cvxpy QP allocation.
Method	Drive Q_{total_target} through a $0 \rightarrow 30$ kVAr ramp.
Sweep	2 policies $\times$ 3 fleet sizes (4, 8, 16) $\times$ 4 capability profiles = 24 runs.
Success	QP improves voltage error metric by ≥ 5 % at the cost of ≤ 50 ms additional dispatch latency.
Artefact	<code>study_C2_q_allocation.png</code> .

C.3 Harmonic-allocation headroom policy (the §11 contribution)

Field	Value
Objective	Validate that headroom-weighted H_{mask} reduces saturation in the most-loaded ICA.
Method	Asymmetric loading of ICAs (one at 90 % S_{max} , others at 30 %).
Sweep	3 imbalance ratios $\times$ 4 harmonic orders compensated = 12 runs.
Success	Saturation events on the loaded ICA reduced by ≥ 40 % relative to equal-share.
Artefact	<code>study_C3_h_allocation.png</code> + table.

C.4 Mode-transition coordination

Field	Value
Objective	Grid-tie open / close transitions across the fleet without any ICA tripping.

Field	Value
Method	Open tie switch at random $t \in [40, 60]$ ms; close at $t \in [120, 140]$ ms.
Sweep	50 Monte-Carlo transitions $\times$ {seamless / staged} schedule = 100 runs.
Success	Zero ICA trips; v_{dc} deviation ≤ 5 % at all ICAs throughout the transition.
Artefact	study_C4_transition_montecarlo.png.

5. Class D — Faults & resilience

D.1 Single-phase grid voltage sag

Field	Value
Objective	Survive a 50 % single-phase sag for 100 ms.
Method	Drop one phase to 0.5 pu in [50, 150] ms; observe FS-MPC and PLL behaviour.
Sweep	3 sag depths $\times$ 3 phases $\times$ 2 PLLs = 18 runs.
Success	No overcurrent trip; SOGI-PLL maintains lock; THD ≤ 8 % during the sag.
Artefact	study_D1_LVRT.png.

D.2 ICA loss-of-comms

Field	Value
Objective	Behaviour when the CMC stops publishing refs.
Method	Stop CMC ticks at $t = 50$ ms; agents fall back on last refs (or local droop in a future extension).
Sweep	1 fault $\times$ 3 fleet sizes = 3 runs.
Success	All ICAs stay online ≥ 5 s past comms loss; THD does not degrade by more than 2 percentage points.
Artefact	study_D2_loss_of_comms.png.

D.3 Sensor-fault injection on one ICA (DT use case U3)

Field	Value
Objective	DT detects a step offset on v_{dc} measurement of one ICA before it destabilises the fleet.
Method	Inject +50 V offset at random $t \in [30, 80]$ ms on one ICA's published v_{dc} .
Sweep	100 Monte-Carlo runs $\times$ {DT on, DT off} = 200 runs.
Success	DT raises critical anomaly within 200 ms of fault onset; rate-of-detection > 95 %.
Artefact	study_D3_dt_sensor_fault.png + ROC curve.

D.4 Cyber attack on Q_{ref} (DT use case U4)

Field	Value
Objective	Cyber screen rejects out-of-bounds and slew-rate violations.
Method	Replay attack: re-publish a 6-month-old Q_{ref} dispatch; bound attack: spoof $Q_{ref} \in \{1 \text{ MVar}, 1 \text{ GVar}\}$.
Sweep	6 attack scenarios.
Success	All attacks detected; alert published within 1 s; per-ICA Q_{ref} either rejected (preferred future) or flagged.
Artefact	<code>study_D4_cyber_attacks.png</code> .

D.5 IGBT short-circuit failure

Field	Value
Objective	Containment when one ICA loses one switch (worst-case device fault).
Method	Force $s_a = 1$ permanently on one ICA at $t = 50$ ms; observe other ICAs and the CMC FDI logic.
Sweep	3 phases $\times$ 4 fleet sizes = 12 runs.
Success	CMC declares fault on the affected ICA within 50 ms; remaining fleet maintains THD ≤ 6 %.
Artefact	<code>study_D5_igbt_fault.png</code> .

6. Class E — Digital-twin validation

E.1 Shadow-plant residual baseline

Field	Value
Objective	Quantify the natural noise floor of the residual under ideal conditions.
Method	DT shadow with identical parameters to the plant; record residuals during 10-cycle steady-state windows.
Sweep	3 modes $\times$ 3 load levels $\times$ 5 trials = 45 runs.
Success	Mean
Artefact	<code>study_E1_residual_baseline.png</code> .

E.2 Phase-locked RLS convergence

Field	Value
Objective	Time-to-converge of <code>PhaseLockedRLS</code> under a real switching profile.
Method	Run 100 cycles of FS-MPC; track $L(k)$, $\hat{r}(k)$; compare against per-sample RLS.
Sweep	3 forgetting factors $\times$ {persistent excitation / quiet} = 6 runs.
Success	PL-RLS within ± 15 % of true (L, r) by cycle 30; per-sample RLS shows $\geq 2x$ variance.
Artefact	<code>study_E2_pl_rls_convergence.png</code> .

E.3 Anomaly-detector ROC

Field	Value
Objective	Receiver-operating-characteristic curve for <code>AnomalyDetector</code> (true-positive vs false-positive).
Method	Inject sensor offsets $\in \{5, 10, 25, 50, 100\}$ V at random times in 1 000-run sample.
Sweep	$5 \text{ offsets} \times \{n_sigma=3, 5, 7\} \times \{dwell=3, 5, 8\} = 45 \text{ configurations} \times 1\,000 \text{ runs}$.
Success	Identify the configuration with $TPR > 0.9$ at $FPR < 0.05$.
Artefact	<code>study_E3_anomaly_roc.png</code> .

E.4 Look-ahead Q-forecaster accuracy

Field	Value
Objective	RMSE of <code>QForecaster.predict()</code> over a 0.2 s horizon vs realised Q.
Method	Subject the load model to a 0.5 Hz sinusoidal Q modulation; record predictions vs actuals.
Sweep	$3 \text{ horizons} \times 3 \text{ modulation amplitudes} = 9 \text{ runs}$.
Success	$RMSE < 10\%$ of peak modulation amplitude.
Artefact	<code>study_E4_q_forecast.png</code> .

7. Class F — Real-time / HIL prep*F.1 Single-tick wall-clock budget*

Field	Value
Objective	Verify that one full FS-MPC tick (predict + cost + outer + telemetry) fits in $20\ \mu\text{s}$ on the target host.
Method	<code>pytest-benchmark</code> on a representative loop; profile the hot path.
Sweep	Three Python builds (CPython 3.11, PyPy 3.10, Numba-jitted).
Success	CPython tick $\leq 80\ \mu\text{s}$, Numba $\leq 5\ \mu\text{s}$ (informs HIL implementation language).
Artefact	<code>study_F1_tick_budget.csv</code> .

F.2 MQTT-broker latency under load

Field	Value
Objective	End-to-end <code>/ica/{id}/refs/*</code> latency at 4-, 8-, 16-ICA scale.
Method	Mosquitto broker + dockerised endpoints; round-trip ping.
Sweep	$3 \text{ fleet sizes} \times \{QoS\ 0, 1, 2\} = 9 \text{ runs}$.
Success	95th-percentile latency $< 10\ \text{ms}$ across all configurations.
Artefact	<code>study_F2_mqtt_latency.png</code> .

F.3 Determinism check (PREEMPT-RT vs vanilla Linux)

Field	Value
Objective	Variance of inner-loop period under sustained load.
Method	30-minute fleet run; record actual T _s histogram.
Sweep	Two kernels (vanilla / PREEMPT-RT) $\times$ 1 fleet size.
Success	PREEMPT-RT 99.9th-percentile jitter $< 5 \mu\text{s}$.
Artefact	study_F3_jitter.png.

8. Aggregate deliverable matrix

Table 8-A — All 25 studies $\rightarrow$ 25 figures + 6 CSV summaries

Class	Study	Figure	CSV	DOF
A	A.1 Three-mode reproduction	loading_mode.png, regenerat- ing_mode.png, statcom_mode.png	THD per mode	0
A	A.2 Energy vs linear PI	study_A2_energy_vs_linear_pi.png	study_A2_parameters.csv	6
A	A.3 Delay compensation	study_A3_delay_compensation.png		9
A	A.4 Load model equivalence	study_A4_load_models.png		3
B	B.1 T _s sweep	study_B1_Ts_surface.png	study_B1.csv	21
B	B.2 L robustness	study_B2_L_robustness.png		12
B	B.3 C robustness	study_B3_C_robustness.png		12
B	B.4 Weak grid	study_B4_weak_grid.png		24
B	B.5 PLL Pareto	study_B5_PLL_pareto.png	study_B5.csv	15
C	C.1 Fleet scaling	study_C1_fleet_scaling.png	study_C1.csv	15
C	C.2 Q allocation	study_C2_q_allocation.png		24
C	C.3 H allocation	study_C3_h_allocation.png		12
C	C.4 Transition Monte-Carlo	study_C4_transition_monte-carlo.png		100
D	D.1 LVRT	study_D1_LVRT.png		18
D	D.2 Loss of comms	study_D2_loss_of_comms.png		3
D	D.3 Sensor fault	study_D3_dt_sensor_fault.png	study_D3.csv	200
D	D.4 Cyber attacks	study_D4_cyber_attacks.png		6
D	D.5 IGBT fault	study_D5_igbt_fault.png		12
E	E.1 Residual baseline	study_E1_residual_baseline.png		45
E	E.2 PL-RLS convergence	study_E2_pl_rls_convergence.png		6
E	E.3 Anomaly ROC	study_E3_anomaly_roc.png	study_E3.csv	45 000
E	E.4 Q forecast accuracy	study_E4_q_forecast.png		9
F	F.1 Tick budget	—	study_F1.csv	3
F	F.2 MQTT latency	study_F2_mqtt_latency.png		9
F	F.3 Jitter	study_F3_jitter.png	—	2

9. Recommended execution order

1. **Class A in week 1** — locks down baselines and confirms the energy-domain trick.
2. **Class B in weeks 2–3** — establishes operating-envelope.
3. **Class C in weeks 4–5** — the multi-ICA story APPEEC paper #2 figures.
4. **Class D in weeks 6–7** — resilience evidence; doubles as IEEE Trans. Smart Grid material.
5. **Class E in weeks 8–9** — DT validation; supports the MAS-DT-SH thesis chapter.
6. **Class F in weeks 10–11** — RT readiness gate before HIL booking.
7. **Week 12** — write-up, figure cleanup, two-pager summary.

The total compute is modest (~360 simulations at ~5 s each $\approx$ 30 minutes single-thread, more with the Monte-Carlo classes). A simple `joblib.Parallel(n_jobs=-1)` makes the whole programme an overnight run.

10. Pre-write-up checklist

Before each paper / chapter draft:

- ☐ All 25 study figures regenerated from the latest commit.
 - ☐ Statistical tests applied where appropriate (paired t-test for A.2; Welch t-test for B.2/B.3).
 - ☐ Confidence intervals on THD numbers reported.
 - ☐ Dependencies frozen in `requirements.lock`.
 - ☐ Random seeds for Monte-Carlo classes (C.4, D.3, E.3) committed.
 - ☐ All figures have legible legends, ≥ 140 dpi, font-size ≥ 9 pt.
 - ☐ Each study's parameters reproducible from a single CLI invocation.
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This plan is the missing piece between Phase 1–3 (which built the system) and Phase 4 (which deploys it). Executing it produces the empirical evidence base for both the APPEEC paper #2 and the MAS-DT-SH thesis Chapters 4–6.